David Nkpa

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Profile

Final-year Computer Science student with hands-on experience building, analysing and pitching AI-enabled solutions across regulated and fast-moving environments. Combines strong technical foundations with structured thinking and clear stakeholder communication to deliver reliable outputs, translate complex concepts for non-technical audiences, and support sound decision-making at pace.

Work Experience

Finance Analyst Apr 2026 – Jun 2026
Rolls-Royce *London, UK*

- Consolidated a 12-month cost dataset, combining 6 months of actuals and forecasts into clean analysis for management review.
- Built an Excel-based model calculating monthly % variances, cumulative spend movement, run-rate trends, and forecast accuracy across departmental cost lines.
- Converted analysis into a 5-slide executive deck, using charts and plain-language commentary to support 3 recommended actions.

Software Engineering Intern Sep 2025 – Oct 2025
Boom Technologies (Fintech) *London, UK*

- Built 10+ admin analytics features in a React-based back-office system, improving operational data visibility and enabling faster, more informed decisions.
- Authored a CEO-ready strategy report on LLM options and sustainable data-centre architectures, evaluating model trade-offs, data governance requirements, and deployment constraints for internal AI initiatives across key markets, documenting technical decisions and risk considerations for non-technical leadership.
- Produced structured technical documentation for proposed AI solutions, covering architecture decisions, integration considerations in line with internal review processes.
- Collaborated across teams to clarify requirements, ship features and communicate trade-offs clearly under time constraints.

Strategy Consultant Intern Jul 2024 – Aug 2024
S&G Strategy *Montreal, Canada*

- Conducted research across 40+ transport providers to evaluate market landscape, pricing models, and competitive positioning.
- Compared leverage and liquidity indicators alongside operating performance to support commercial recommendations.

Projects

Live DeFi Liquidity Agent Jun 2026
Python, Hidden Markov Models, NumPy

- Won 1st place in the TurinTech AI track at UKFinnovator London Hackathon 2026, Imperial College London; pitched the final solution to industry judges and challenge setters from King's College London and TurinTech AI.
- Built a Hidden Markov Model agent in pure NumPy detecting three market regimes in real time via Bayesian posterior updates to manage concentrated liquidity in a Uniswap V3-style AMM simulation.
- Engineered dwell-based rebalancing with directional band skewing, triggering only when profit exceeded gas cost; outperformed all provided baselines on hidden evaluation seeds.

AI Financial Decision Assistant (Final-Year Project) Sep 2025 – May 2026
Python, FastAPI, LLMs

- Built a full-stack web app for retail investors to analyse US stocks using risk/return metrics, including volatility, drawdown and decision-ready summaries.
- Created a Python/FastAPI analytics service to ingest market data and compute indicators with Pandas/NumPy, including RESTful API integration with external financial data sources, with robust error handling for consistent outputs.

- Designed an agentic LLM layer using prompt engineering, output constraints, and tool-use patterns with hosted APIs (including Claude and OpenAI) to generate plain-language stock comparisons, applying responsible AI principles to ensure outputs remain grounded and auditable.

Metropolitan Police Crime Prediction Model Mar 2026

Python, Pandas, Scikit-learn, Random Forest, K-Means, MapReduce

- Built a predictive analytics pipeline on 1.1M+ Metropolitan Police crime records, cleaning and aggregating incident-level data into 56,571 neighbourhood-month modelling rows.
- Engineered temporal, geographic and offence-composition features, including lagged crime counts, crime-type ratios and Shannon entropy, while applying leakage checks and time-based train-test splitting.
- Compared Linear Regression and Random Forest models, achieving 0.934 test R^2 with Random Forest and identifying previous-month crime count as the strongest predictor.
- Applied K-Means clustering and MapReduce-style processing to profile neighbourhood crime patterns and support resource-allocation insights.

AWS Cloud Architecture & Security Design - Private Cloud on AWS Nov 2025 – Dec 2025

AWS, EKS, Kubernetes

- Designed and implemented a cloud-native, fault-tolerant architecture on AWS using Kubernetes (EKS), supporting horizontal scaling and self-healing workloads.
- Engineered network isolation using VPCs, public/private subnets, and NAT gateways, aligning with cloud security best practices.
- Deployed stateless application workloads behind managed load balancers, improving resilience and reducing blast radius.
- Implemented shared state via Amazon ElastiCache (Redis) with consistency-first behaviour under network partitions, explicitly reasoning about CAP trade-offs.

Education

University of Leicester Leicester, UK

BSc Computer Science *Sep 2023 – Jun 2026*

- Awarded First-Class Honours (Bachelor of Science with Honours in Computer Science).
- Citizens of Change 100 Scholarship Winner (2023).
- **Relevant modules:** Big Data and Predictive Analytics; Data Analytics; Databases and Domain Modelling; Computers, Society and Professionalism, Introduction to Cloud Computing

The British School of Lomé Lomé, Togo

IGCSE (8 subjects: 7 A and 1 A, including Math, ICT, and Physics)* *Aug 2017 – Jun 2021*

- Duke of Edinburgh (Bronze) Award.

Skills

- **Work quality & operations:** attention to detail, documentation
- **Writing & reporting:** Microsoft 365 (Word, Excel, PowerPoint), Canva, Figma
- **Technical:** Python (Pandas/NumPy), React.js, SQL, HTML/CSS, REST APIs
- **AI:** Generative AI/LLMs, prompt engineering, agents, Claude, OpenAI APIs

Leadership

Course Representative Oct 2025 – Present

School of Computing & Mathematics, University of Leicester *Leicester, UK*

- Represent 200+ students in departmental meetings; gather requirements, resolve issues, and communicate outcomes clearly.
- Synthesised cohort feedback into actionable themes for staff, improving communication and follow-through.

Languages

- **English:** Native
- **French:** Working Proficiency